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Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2025

Code: 101306 ***Time: 03 Hours***

Course: B.Tech ***Subject: Materials, Testing & Evaluation*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) The main compound responsible for early strength of cement is
(i) C2S (iii) C3A
(ii) C3S (iv) C4AF

(b) Graphene is known for
(i) Poor electrical conductivity (iii) High strength and conductivity
(ii) Brittleness only (iv) High porosity

(c) True stress is calculated using
(i) Original area (iii) Reduced instantaneous area
(ii) Average area (iv) Gauge length

(d) Hooke's law is valid up to
(i) Yield point (iii) Fracture point
(ii) Elastic limit (iv) Ultimate load

(e) In a tensile test, ductility is measured by
(i) Hardness number (iii) Modulus of elasticity
(ii) % elongation (iv) Poisson's ratio

(f) Creep becomes significant in metals at
(i) Low temperature only (iii) Elevated temperature under constant stress
(ii) Room temperature only (iv) Zero stress

(g) Fatigue failure occurs due to
(i) Static overload (iii) Repeated cyclic loading
(ii) Corrosion only (iv) Compression only

(h) The transition temperature phenomenon is mainly associated with
(i) Aluminum (iii) Copper
(ii) Mild steel (iv) Rubber

(i) Fracture toughness indicates resistance to
(i) Wear (iii) Crack propagation
(ii) Corrosion (iv) Thermal expansion

(j) The modulus of resilience is the energy absorbed within
(i) Plastic range (iii) Elastic range
(ii) Creep zone (iv) Necking zone

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Q.2 Solve both questions : **[14]**

(a) Analyze the role of major cement compounds in strength development and setting behavior of cement.

(b) Evaluate the suitability of high-performance concrete and lightweight concrete for high-rise structures.

Q.3 Solve both questions : **[14]**

(a) What is fatigue failure? Explain S-N curve and factors affecting fatigue life.

(b) Evaluate the effectiveness of geo-textiles in soil stabilization, filtration and drainage systems.

Q.4 Solve both questions : **[14]**

(a) Examine the functional requirements of paints and varnishes for protection of steel structures in aggressive environments.

(b) Explain plastic deformation of metals and strain hardening phenomenon.

Q.5 Solve both questions : **[14]**

(a) Explain and evaluate the tensile testing procedure for brittle, quasi-brittle and ductile materials.

(b) Compare and evaluate fracture toughness of brittle and ductile materials.

Q.6 Solve both questions : **[14]**

(a) What are acoustical materials? Discuss their characteristics and uses in buildings.

(b) Describe properties and engineering applications of ceramics and refractories.

Q.7 Solve both questions : **[14]**

(a) Evaluate the role of fracture mechanics in preventing catastrophic structural failures.

(b) What is M-Sand? Compare M-Sand with natural river sand.

Q.8 Solve both questions : **[14]**

(a) Compare and justify the selection of Brinell, Rockwell and Vickers hardness tests for different materials.

(b) Discuss engineering properties and future applications of graphene and carbon composites.

Q.9 Write short note on any two of the following: **[7 x 2 = 14]**

(a) Bitumen and asphaltic materials used in pavement construction

(b) Properties of lightweight concrete and high-performance concrete

(c) Rubber, asbestos, laminates and adhesives

(d) Engineering stress-strain curve and true stress-strain curve

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